

SEMESTER-VI

UMT802: INDUSTRIAL AUTOMATION				
	L	T	P	Cr
	3	0	2	4.0
Course objectives: This course focuses on understanding the various components of state-of-the-art automation technologies encountered in modern manufacturing industries. Also, introduces the practical methods of automatic control of machines, processes, and systems through various automation approaches such as pneumatic control, electropneumatic relay logic control, programmable logic controller, and IIoT.				
Syllabus: Need of Automation and Integration: Introduction to automation technologies, strategies, applications around us and in mechanical industries. Types of systems - mechanical, electrical, electronics. Hydraulic and Pneumatic Systems: Elements of hydraulic/pneumatic system, fluid power control elements and standard graphical symbols for them, hydraulic and pneumatic cylinders, hydraulic pumps, hydraulic and pneumatic valves for pressure, flow and direction control, safety valves. Circuit Design Approach for Hydraulic and Pneumatic Control Systems: Design and operation of logic control circuits for real time examples; sequence and synchronize the operation of two or more than two cylinders as per the design requirements to automate the various systems in modern industries and equipment. Hydraulic and pneumatic safety and their applications to clamping, traversing, and releasing operations. Hydraulic system analysis. Design and Operation of Electro-Pneumatic Control Circuits: Elements of electro-pneumatic, advantages over hydraulic and pneumatic control, solenoid valves, relays, factory automation sensors, and their interfaces. Electro-pneumatic control circuits design using relay logic approach and examples, sequence control of a lifting device, sequence and synchronize the operation of two or more than two cylinders/actuators, electro-pneumatic systems, feedback control systems. Industrial Control Systems: Programmable Logic Controller (PLC), advantages over electropneumatic relay logic control systems, programming languages & instruction sets for ladder logic, and their applications with various automation examples in manufacturing industries, various types of time delay and counting operations. Electrical connection diagrams or interfacing of various input and output devices with the PLC. Human Machine Interface (HMI) & Supervisory Control and Data Acquisition System (SCADA), motion controller, smart sensors, RFID technology and its application, machine vision and control applications.				
Laboratory Work Hands-on logic circuits design and operation of automatic systems for various industrial applications. Pneumatic logic circuits design and simulation using FluidSIM software as well as implementation on pneumatic trainer kit for checking the functional applications. Electro-pneumatic relay logic circuits design and simulation using FluidSIM software as well as implementation on electro-pneumatic trainer kit for checking the functional applications. Programmable logic controller (PLC) – Ladder logic programming using CoDeSys V2.3 or RS Logix 500 software, interfacing of different input switches or sensors and output solenoids with PLC, and implementation for different industrial automation				

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applications. Use of CIROS software for 3D simulation of various automation technologies in a virtual environment. Use of Industrial Internet of Things (IIoT).

Course Learning Outcomes (CLOs)

The students will be able to

1. recognize the benefits and applications of automation technologies in various contemporary manufacturing systems
2. design a hydraulic system or process after identifying the various control valves, actuators, and other related components to meet the desired needs within realistic constraints for contemporary industrial systems.
3. design pneumatic logic circuits or electro-pneumatic logic circuits for various process control applications in industry.
4. apply the PLC automation approach to control the complex industrial processes necessary for contemporary engineering practice.

Text Books

1. Automation Production System & Computer Integrated Manufacturing, Groover, M., Pearson Education Asia, 2009.
2. Fluid Power with Applications, Esposito, A., Pearson Education 6th ed., 2009.
3. Pneumatic Systems, Majumdar, S. R., McGraw Hill, 2005.
4. Programmable Logic Controllers – Programming Methods and Applications, John R. Hackworth & Frederick D. Hackworth Jr, Pearson, 2011.
5. Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications, Minoli, D., Wiley, 2013.

Reference Books

1. Theory and Applications of Automatic Controls, Nakra, B. C., New Age International Publishers, Revised 2nd ed., 2014.
2. Automated Manufacturing Systems, Morriss, S. B., McGraw Hill, 2006.
3. Mechatronics: Mechanical System Interfacing, Auslander, D. M. and Kempf, C. J., Prentice Hall Inc., New Jersey, 1996.
4. Programmable Logic Controllers – Principles and Applications, John W. Webb & Ronald A. Reis, Pearson Education, 5th ed., 2008.
5. FESTO Workbook of Pneumatic, Electro-pneumatics and PLC.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-40
3	Sessional (Assignments/Projects/Quizzes/Lab Evaluations/Lab Reports)	30

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